

# Computing for Analytical Work

**ECBS5293 · DS1 · MSBA 2026–2027**

**Before Block 1: what this course is, who we are, how it is graded**

## What this course is — and is not

- **One credit.** Three Mondays, 13:30 to 17:20 each, and one closed-book exam in reading week, about four weeks after the last one.
- **Not a Python course.** The layer *beneath* your other courses: the filesystem, the shell, processes, Python environments, Git — and the habit of diagnosing what actually went wrong.
- **Three sessions, one question each:** Where am I? · What is running? · What broke?
- **Every lab is a broken project.** You diagnose it, repair it, verify the repair, and explain it. The explanation is the skill.

## Who I am, as your professor

- **Eduardo Ariño de la Rubia** — Professor of Practice in Data Science and AI, Department of Economics and Business. You may meet me again in Data Science 2 and in Deep Learning and Applied AI.
- **Industry, before this:** Meta, Senior Director of Data Science, Reality Labs · Domino Data Lab, Chief Data Scientist and VP of Product · Ingram Content Group, Principal Data Scientist and robotics lead.
- **Elsewhere:** co-founded DataScience.LA (1,000+ members) · talks at KDD, rstudio::conf, and Rev · host of CEU's *Teaching Analytics in the Age of AI* podcast.
- **Also trained in negotiation and conflict resolution** — which is why, in a lab, you get questions from me, not answers.
- [RubiaE@ceu.edu](mailto:RubiaE@ceu.edu) · office hours by appointment · GitHub `earino`

## Who else is here

- **Boga** — teaching assistant, in the room every session. During labs: questions, not answers.
- **Nahian** — teaching assistant, remote. Reads every lab checkpoint after class and replies on Moodle to every *need help* line before the next session.
- Each of us has been stuck on exactly the kind of thing this course teaches. Thirty seconds each.

## How a session runs — 13:30 to 17:20

|               |                                                    |
|---------------|----------------------------------------------------|
| 13:30 – 15:10 | Block 1: 45 lecture · 5 stand-and-stretch · 45 lab |
| 15:10 – 15:40 | break – thirty minutes, leave the room             |
| 15:40 – 17:20 | Block 2: 45 lecture · 5 stand-and-stretch · 45 lab |

- Sessions 2 and 3 open with a **10-minute closed-book knowledge check**, out of that block's lecture, never the lab.
- **The lab is the course.** Every lab ends the same way: you **explain it to a neighbour**, hand up if unsure, and upload your diagnosis note to the lab's **Moodle checkpoint** before you leave. That is what complete means. Block 1 shows you the details.
- **Labs carry no points.** They cost you preparation if skipped, not marks.
- Bring the laptop every session. Windows: **Git Bash**, always.

## Three Mondays, one arc

| Session | Date       | Question                                                          | Homework                  |
|---------|------------|-------------------------------------------------------------------|---------------------------|
| 1       | Mon 14 Sep | Where am I? — files, paths, the terminal, <code>git status</code> | HW1 due Fri 18 Sep, 23:59 |
| 2       | Mon 21 Sep | What is running? — interpreters, environments, notebooks, commits | HW2 due Fri 25 Sep, 23:59 |
| 3       | Mon 28 Sep | What broke? — tracebacks, data files, recovery                    | HW3 due Fri 9 Oct, 23:59  |

**Final exam:** closed-book, in person, about 70 minutes, in **reading week**, about four weeks after Session 3.

The date is on Moodle. A **mock exam** with a key comes first.

Under all of it, one loop: **observe → locate → hypothesize → inspect → change → verify → explain.**

## How you are graded

| Component                                                                   | Weight |
|-----------------------------------------------------------------------------|--------|
| Homework 1 — Files, paths, terminal, and run instructions                   | 10%    |
| Homework 2 — Python execution, environments, notebooks, and Git checkpoints | 15%    |
| Homework 3 — Debugging, data files, recovery, and final diagnosis note      | 15%    |
| Knowledge checks (start of Sessions 2 and 3, closed-book)                   | 10%    |
| Final exam (closed-book, in person, in reading week)                        | 50%    |

**Homework** = repair a broken repository + a written diagnosis note + a short video walkthrough in your own voice, graded against a rubric published with the assignment. The explanation carries more weight than the fix. A self-check key posts on Moodle after each late window closes.

## The rules that will actually matter to you

- **Late homework:** one day late at –10%; nothing after Saturday 23:59, because the key posts Sunday.
- **Extensions:** email me *before* the deadline, with a reason. Documented illness or emergencies are always accommodated.
- **Missed knowledge check:** documented absence → that 5% moves to the final. Undocumented → 0. No make-ups.
- **Missed lab:** do it at home that week and submit the checkpoint; say in the note if you want someone to look.
- **Regrading:** within 7 days, in writing, naming the rubric criterion. The whole submission is re-read; the grade can move either way.
- **Accommodations:** the skill is diagnosis, not speech. Written explanations and written or audio walkthroughs exist — arrange them with me in advance.
- **Moodle is authoritative** for every date.

## AI: use it to learn, not to avoid learning

- **Yes:** explain this error; suggest what to inspect; teach me the concept; improve my writing.
- **No:** submit code or explanations you cannot defend in person; run commands you cannot explain; describe work you did not do.
- The grading **assumes** you used AI. It checks whether you understood the result: the video, the neighbour, the closed-book exam.
- **Labs 3 and 5 open with 15 minutes of no AI.** Terminal and error text only. Announced now, so it is never a trap.
- You are responsible for everything you run, change, and submit.

## Before Block 1

- **Two places:** the course site — syllabus, setup, glossary — at `earino.github.io/ecbs5293` , and **Moodle** — deadlines, checkpoints, submissions. Labs and homework are public GitHub repos you clone. You never push. No GitHub account needed.
- **Setup check.** If yours did not end in `ALL CHECKS PASSED` , or you have not run it: tell Boga or me **now**. We come to you during the lab, not the lecture.
- **The glossary** is for you. A word on the site that confuses you is a bug in the page, not in you. Tell a TA.
- Laptops closed for the first ten minutes of Block 1.